EBN111 Formulas

Chapter 1

Factor	Prefix	Symbol
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

Quantity	Unit	Symbol
electric charge	coulomb	C
electric potential	volt	V
resistance	ohm	Ω
conductance	siemens	S
inductance	henry	H
capacitance	farad	F
frequency	hertz	Hz
force	newton	N
energy, work	joule	J
power	watt	W
magnetic flux	weber	Wb
magnetic fiux density	tesla	T

e = -1.602×10^{-19} C

$$Q = \int_{t0}^{t1} i\,dt$$

i = dq/dt

1A = 1C/s

V_{ab} = dw/dq

$V_{ab} = V_a - V_b$

p = $\frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt}$ = vi

1 Wh = 3600 J = 3.6 kJ

Chapter 2

R = $\rho \frac{l}{Area} = \frac{V}{I}$

V = IR

G = $\frac{1}{R}$ or $i = Gv$

$b = l + n - 1$

b branches, **n** nodes, and **l** independent loops

KCL:

$$\sum_{n=1}^{N} i_n = 0$$

KVL:

$$\sum_{m=1}^{M} V_m = 0$$

Voltage Division:

$$V_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} V$$

Current Division (only 2):

$$I_n = \frac{R_{n\_other}}{R_1 + R_2} I$$

Current Division (any):

$$I_n = \frac{G_n}{G_1 + G_2 + \cdots + G_N} I$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \qquad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

Delta ⟶ Wye:

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

Wye ⟶ Delta:

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_4}{R_1}$$

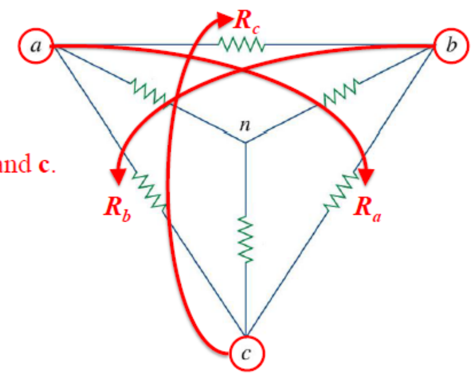
$$R_a = R_b = R_c = R_\Delta$$

and

$$R_1 = R_2 = R_3 = R_Y$$

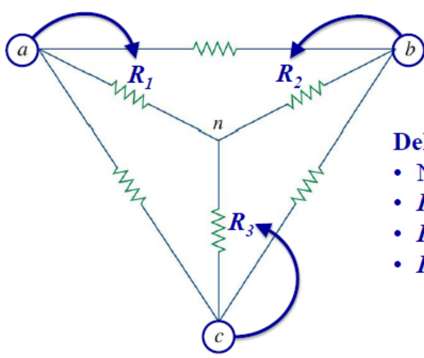
then

$$R_Y = \frac{R_\Delta}{3} \; or \; R_\Delta = 3R_Y$$



Delta Δ: R_a, R_b, R_c.

- Name the nodes **a**, **b**, and **c**.
- ***R_a*** is opposite node **a**
- ***R_b*** is opposite node **b**
- ***R_c*** is opposite node **c**



Delta Y: R_1, R_2, R_3.

- Name the nodes **a**, **b**, and **c**.
- ***R_1*** is connected to node **a**
- ***R_2*** is connected to node **b**
- ***R_3*** is connected to node **c**

Chapter 3

Nodal analysis:

$$i = \frac{V_{High} - V_{Low}}{R_{Branch}} \qquad i_{Branch} = \frac{V_{n\_from} - V_{n\_to}}{R_{Branch}}$$

Mesh analysis:

$$\mathrm{R}(i_{\mathrm{loop}} - i_{opposing\_loop})$$

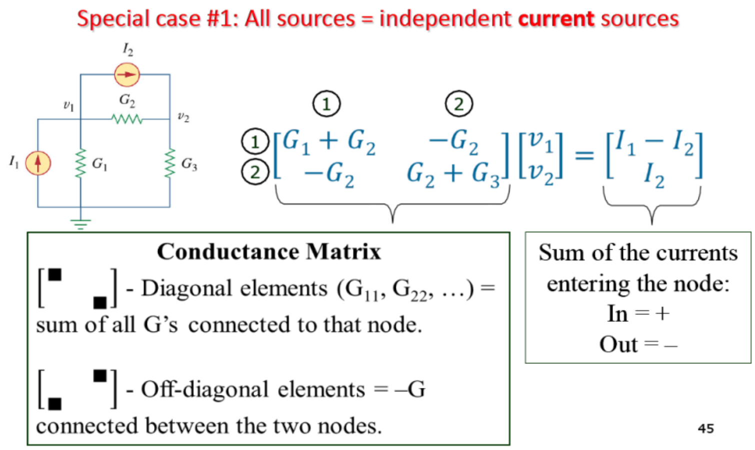
Transistors:

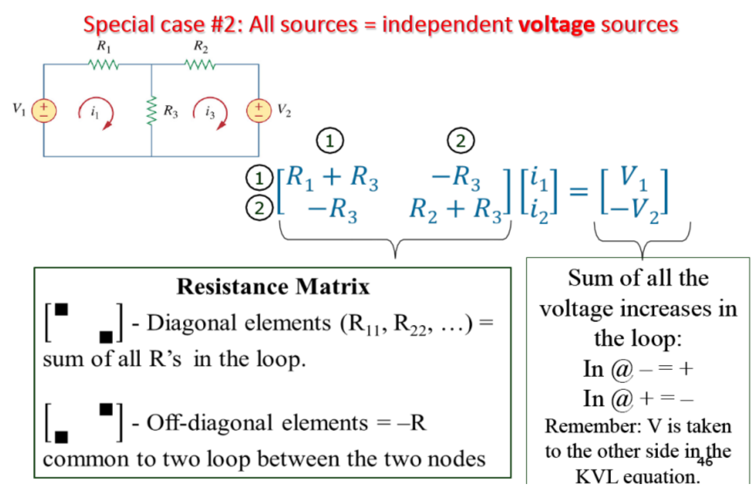
$$V_{CE} = V_{BE} + V_{CB} \qquad I_E = I_C + I_B$$

$$V_{BE} = 0.7V \qquad I_C = \beta I_B$$

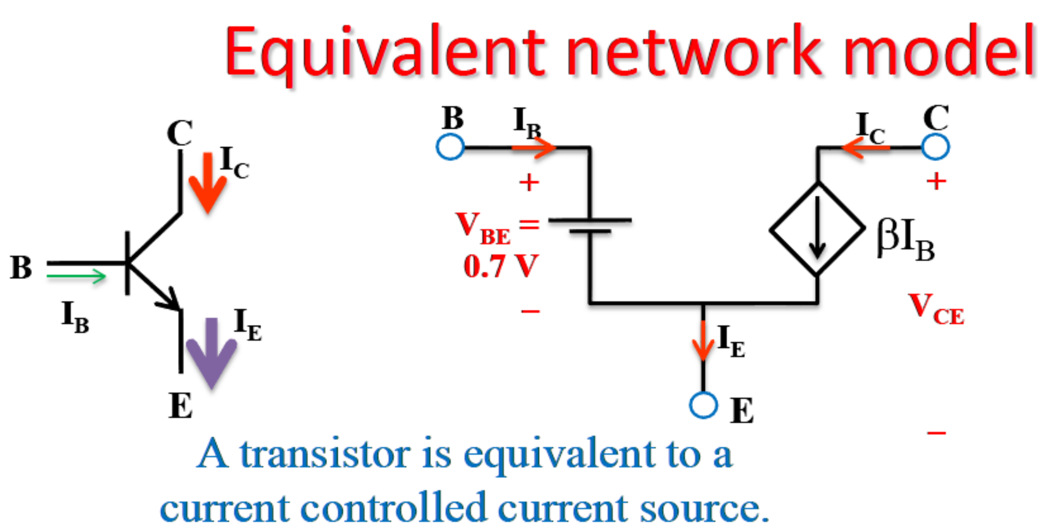
$$\beta = \frac{\alpha}{1 - \alpha} \qquad I_E = (1 + \beta) I_B$$

Nodal and Mesh Analysis by inspection:





Transistors:



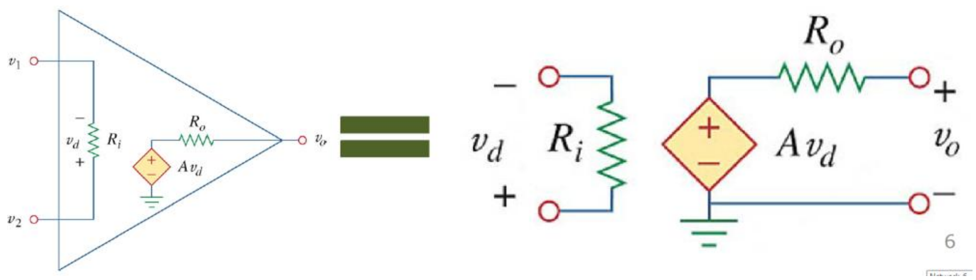
Chapter 4

$$P = i^2 R_L = \left(\frac{V_{TH}}{R_{TH} + R_L}\right)^2 R_L$$

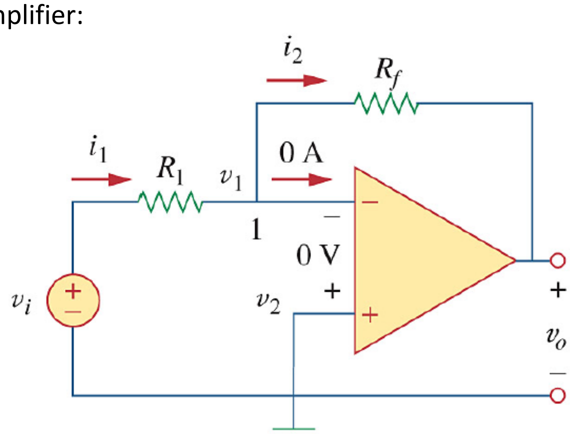
For max power: $R_L = R_{TH}$

$$P_{max} = \frac{{V_{TH}}^2}{4R_{TH}} = \frac{{V_{RL}}^2}{R_{TH}}$$

Chapter 5



Inverting Amplifier:



$v_+ = v_- = 0\text{V}$

KCL at v-

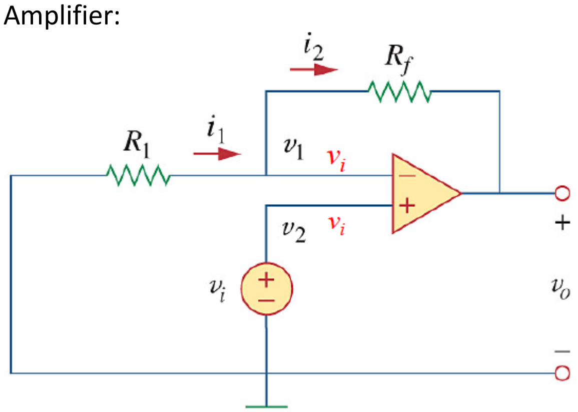
$$-i_1 + i_2 = 0$$

$$\frac{0 - V_i}{R_1} + \frac{0 - V_o}{R_f} = 0$$

$$\frac{V_o}{V_i} = -\frac{R_f}{R_1}$$

$$\boxed{V_o = -\frac{R_f}{R_1} V_i}$$

Non-inverting Amplifier:



$v_+ = v_- = V_i$

KCL at v-

$$\frac{V_i - 0}{R_1} + \frac{V_i - V_o}{R_f} = 0$$

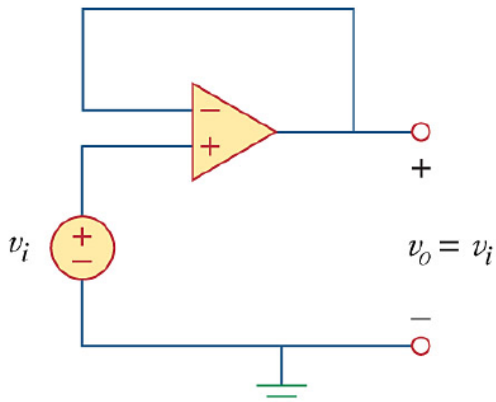
$$\frac{V_i - V_o}{V_i} = -\frac{R_f}{R_1}$$

$$1 - \frac{V_o}{V_i} = -\frac{R_f}{R_1}$$

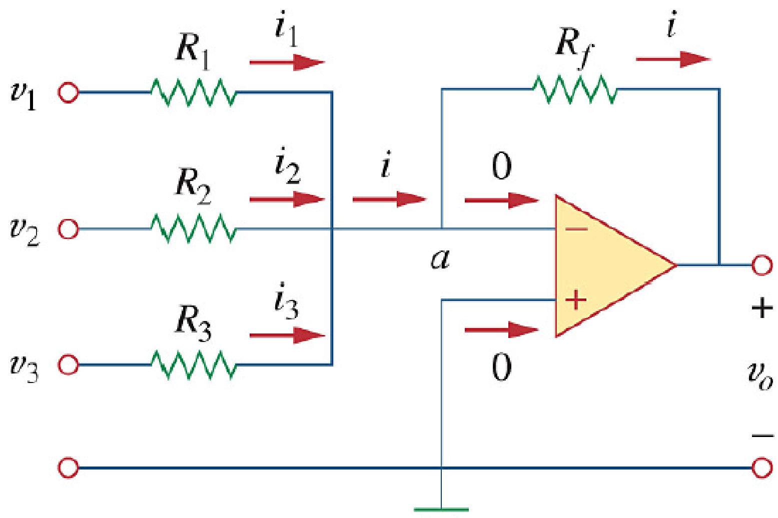
$$\frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}$$

$$\boxed{V_o = \left(1 + \frac{R_f}{R_1}\right) V_i}$$

Buffer:

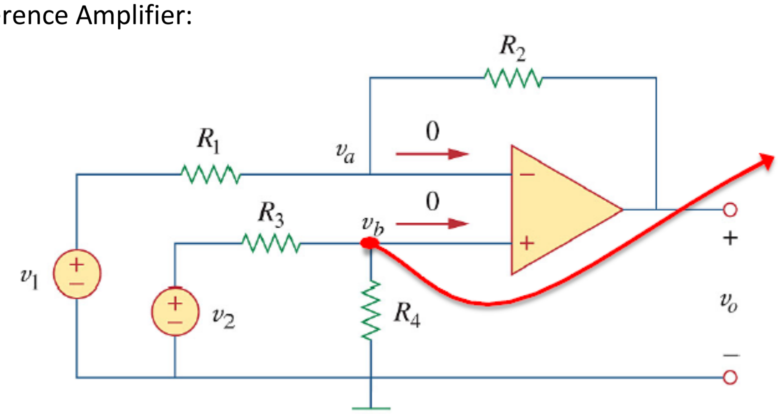


Summing Amplifier:



$$V_o = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$$

Difference Amplifier:



$$v_b = \frac{R_4}{R_3 + R_4}v_2 = v_a$$

See derivation from 1st principles in textbook p187 and p188

$$\boxed{v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)}v_2 - \frac{R_2}{R_1}v_1}$$

1. If $R_1/R_2 = R_3/R_4$: $V_o = 0$ if $V_1 = V_2$
2. If $R_2 = R_1$ and $R_3 = R_4$: $V_o = V_2 - V_1$

Chapter 6

Capacitors:

$$C = \frac{\varepsilon A}{d} \qquad [F]$$

$$C = \frac{q}{v}$$

$$i = C\frac{dv}{dt}$$

$$v = \frac{1}{C}\int_{t_0}^{t} i(t)\,dt + v(t_0)$$

$$w = \frac{1}{2}Cv^2 = \frac{q^2}{2C}$$

$$p = vi = Cv\frac{dv}{dt}$$

Series: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_N}$

Parallel: $C_{eq} = C_1 + C_2 + \cdots + C_N$

Inductors:

$$L = \frac{N^2\,\mu\,A}{l} \qquad [H]$$

$$v = L\frac{di}{dt}$$

$$i = \frac{1}{L}\int_{t_0}^{t} v(t)\,dt + i(t_0)$$

$$w = \frac{1}{2}Li^2$$

$$p = vi = L\,\frac{di}{dt}\,i$$

Series: $L_{eq} = L_1 + L_2 + \cdots + L_N$

Parallel: $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_N}$

Chapter 9

$$v(t) = V_m \sin(\omega t + \varphi) \qquad T = \frac{2\pi}{\omega}$$

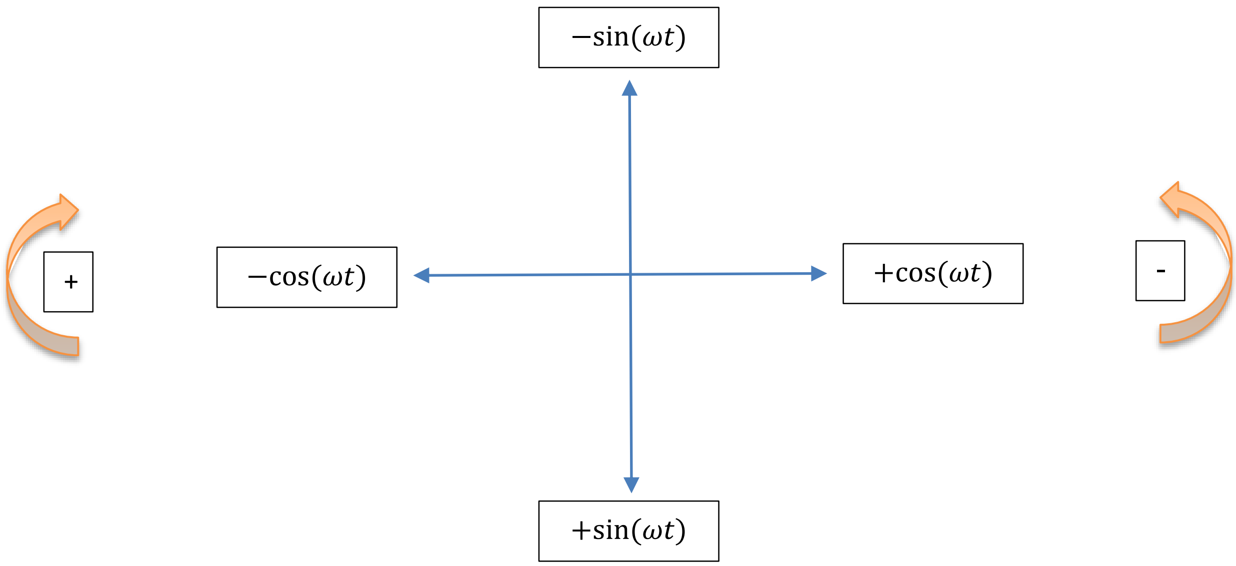
$$f = \frac{1}{T} \qquad \omega = 2\pi f$$

$$\sin(\omega t \pm 180°) = -\sin(\omega t)$$
$$\cos(\omega t \pm 180°) = -\cos(\omega t)$$

$$\sin(\omega t \pm 90°) = \pm\cos(\omega t)$$
$$\cos(\omega t \pm 90°) = \mp\sin(\omega t)$$

Rectangular: $z = x + jy = r(cos\varphi + jsin\varphi)$
Polar: $z = r\angle\,\varphi°$
Exponential: $z = re^{j\varphi}$

$$\frac{1}{j} = -j$$
$$r = \sqrt{x^2 + y^2}$$
$$\varphi = \tan^{-1}\left(\frac{y}{x}\right)$$



Arithmetic:

Addition (+): MUST WORK IN RECTANGULAR FORMAT

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction (–): MUST WORK IN RECTANGULAR FORMAT

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication (×): WORK IN POLAR FORMAT

$$z_1 z_2 = r_1\angle\varphi_1{}^\circ . r_2\angle\varphi_2{}^\circ = r_1 r_2\angle(\varphi_1{}^\circ + \varphi_2{}^\circ)$$

Division (÷): WORK IN POLAR FORMAT

$$\frac{z_1}{z_2} = (r_1\angle\varphi_1{}^\circ)/(r_2\angle\varphi_2{}^\circ) = \frac{r_1}{r_2}\angle(\varphi_1{}^\circ - \varphi_2{}^\circ)$$

Root ($\sqrt{}$): $\quad \sqrt{z} = \sqrt{r}\angle\left(\frac{\varphi}{2}\right)$

Inverse $\left(\frac{1}{z}\right)$: $\quad \frac{1}{z} = \frac{1}{r}\angle(-\varphi)$

Conjugate (z^*): $\quad z^* = (x + jy)^* = (x - jy)$

$$= r\angle(-\varphi)$$

$$= e^{-j\varphi}$$

Euler's identity: $\quad e^{\pm j\varphi} = cos\varphi \pm j\, sin\varphi$

(time domain) $\leftrightarrow$ (phasor domain)

$$v(t) = V_m \cos(\omega t + \varphi) \leftrightarrow \bar{V} = V_m\angle\varphi$$

$$i(t) = I_m \cos(\omega t + \varphi) \leftrightarrow \bar{I} = I_m\angle\varphi$$

Element	Time domain	Frequency domain	Impedance	Admittance
R	$v = Ri$	$\bar{V} = R\bar{I}$	$Z = R$	$Y = \dfrac{1}{R}$
L	$v = L\dfrac{di}{dt}$	$V = j\omega LI$	$Z = j\omega L$	$Y = \dfrac{1}{j\omega L}$
C	$i = C\dfrac{dv}{dt}$	$V = \dfrac{I}{j\omega C}$	$Z = \dfrac{1}{j\omega C}$	$Y = j\omega C$

L:	DC	Short circuit	$\omega = 0, \quad Z = 0$
	High f	Open circuit	$\omega = \infty, \; Z = \infty$

C:	DC	Open circuit	$\omega = 0, \quad Z = \infty$
	High f	Short circuit	$\omega = \infty, \; Z = 0$